

Prashant Agrawal

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EDUCATION

Indian Institute of Information Technology, Allahabad

B.Tech, Electronics and Communication Engineering, CGPA: 8.57

Prayagraj, Uttar Pradesh

Nov. 2022 - Present

EXPERIENCE

Locus.sh

Jan 2026 - Present

Software Engineer Intern

- Owned end-to-end a scalable, config-driven automation pipeline (**Node.js**, **TypeScript**, **AWS**) for client onboarding on a **1B+ delivery** platform, cutting a **4+ hour** manual workflow to a single API call.
- Improved pipeline performance by **~90%** by designing a modular architecture with parallel entity migration, dependency-grouped execution, and **idempotent upserts** with change detection.
- Built a real-time monitoring dashboard using **Server-Sent Events (SSE)** with per-user log isolation, using telemetry and metrics to track live pipeline activity and drive operational excellence.

Bindisa Agritech

May 2025 - Jul 2025

Software Engineer Intern

- Developed a real-time observability platform using **Python**, **Streamlit**, and **Prometheus** to track latency and error metrics across microservices, reducing debugging cycles by **25%**.
- Designed a self-service tool that lets engineers manage service mesh policies through a UI without writing code, eliminating manual configuration work for the platform team.

PROJECTS

Self-Healing Distributed System

🔗 GitHub

Tech Stack: Python, FastAPI, Docker, AWS (Lambda, EventBridge, S3, EC2)

- Architected a scalable, event-driven system using **FastAPI**, **Docker**, and **AWS** that automatically detects and recovers failed microservices in **5-30 seconds** without manual intervention.
- Designed a severity-aware recovery engine with **circuit breaker** and per-service strategies (restart, fallback, escalate), using sliding-window failure tracking to escalate from LOW to CRITICAL automatically.
- Built a test strategy of **74 automated unit and integration tests** covering circuit-breaker transitions and simulated failures; deployed on **AWS EC2** with a **CloudWatch dashboard** and **S3** crash reports.

WindowShop_AI (HackOn With Amazon)

🔗 GitHub 🌐 Live

Tech Stack: Flask, JavaScript, PyTorch, OpenCV, YOLOv8, CLIP

- Led a team of 3 to build **WindowShop_AI**, a real-time in-video product discovery tool; designed the **Flask** backend and REST APIs to orchestrate the full detection-to-recommendation pipeline.
- Built an AI pipeline using **YOLOv8** & **CLIP** achieving **92% identification accuracy**; optimized latency by pre-computing and caching product embeddings in **PyTorch**.

GENAI Smart Medical Chatbot

🔗 GitHub 🌐 Live

Tech Stack: Flask, LangChain, Llama, Pinecone, HuggingFace, SQLite

- Built a **RAG-based medical chatbot** using **LangChain**, **Llama 3.1**, and **Pinecone**, with **SQLite**-backed conversation memory for context-aware multi-turn interactions.
- Designed retrieval pipeline with text chunking, **384-dim HuggingFace embeddings**, and top-k semantic search for accurate document retrieval.

TECHNICAL SKILLS

- Languages:** C++, Python, JavaScript, TypeScript, SQL
- Backend:** Node.js, Express.js, Flask, FastAPI, REST APIs
- Testing & Tools:** Unit & Functional Testing, Automated Test Suites, Git, New Relic, Prometheus
- DB & Infra:** MySQL, MongoDB, Docker, Kubernetes, AWS, Linux, CI/CD, WebSockets
- Frontend & Core:** React.js, HTML/CSS; DSA, System Design, OS, CN, OOP

ACHIEVEMENTS & CODING PROFILES

- LeetCode:** archief_117 (Knight) | **Codeforces:** goldrevboy (Expert) | **CodeChef:** aa333 (5★ Star) | **1000+** problems solved across platforms.
- HackOn with Amazon:** National Semifinalist - Top 10 teams | **Flipkart Grid:** National Semifinalist - Top 30 teams.